

# HCLTech - 100 Technical MCQ Questions

## HCL Technologies

For BTech Computer Science / IT Students

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### Object-Oriented Programming

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**1. Which concept allows objects of different classes to be treated as objects of a common superclass?**

- A) Polymorphism
- B) Encapsulation
- C) Abstraction
- D) Overloading

**2. What is a virtual function used for in C++?**

- A) To achieve runtime polymorphism
- B) To avoid inheritance
- C) To create static memory
- D) To improve compilation speed

**3. Which of the following is true regarding static methods and polymorphism?**

- A) Static methods support runtime polymorphism
- B) Static methods are resolved at compile time and do not support runtime polymorphism
- C) Static methods cannot be called
- D) Static methods must be abstract

**4. Constructor overloading is possible when:**

- A) Constructors have same name but different parameter lists
- B) Constructors are private
- C) A class has no constructor
- D) Only one constructor exists

**5. What happens if a class does not implement all abstract methods of an interface it implements?**

- A) Compilation succeeds silently
- B) The class must be declared abstract
- C) Runtime error occurs
- D) Nothing happens

**6. Which access modifier makes a member accessible only within the same class?**

- A) public
- B) protected
- C) private
- D) default

**7. A class that cannot be instantiated directly is called a(n):**

- A) Abstract class
- B) Final class
- C) Static class
- D) Sealed class

**8. What is a 'friend' function in C++?**

- A) A function that has access to private/protected members of a class
- B) A public function only
- C) A function that cannot access class members
- D) A default constructor

**9. Which of these is an example of compile-time polymorphism?**

- A) Method overriding
- B) Operator overloading (in C++)
- C) Dynamic method dispatch
- D) Runtime type identification

**10. Which of these is true about interfaces in Java 8+?**

- A) They can have default and static methods with bodies
- B) They can have instance variables with values changed at runtime
- C) They support constructors
- D) They cannot be implemented by multiple classes

**11. What is method hiding in OOP?**

- A) When a static method in a subclass has the same signature as one in the superclass
- B) When a private method is called
- C) When a method throws an exception
- D) When constructors are overloaded

**12. Can a constructor be overridden?**

- A) Yes, always
- B) No, constructors cannot be overridden
- C) Only static constructors can
- D) Only in interfaces

**13. In Java, which keyword is used to inherit a class?**

- A) implements
- B) extends
- C) inherits
- D) super

**14. Which of the following best describes 'abstraction'?**

- A) Showing only relevant details and hiding implementation complexity
- B) Combining data and methods
- C) Creating multiple objects
- D) Executing code faster

**15. Which design principle states 'favor composition over inheritance'?**

- A) SOLID-related OOP design guideline
- B) A rule that disallows interfaces
- C) A Java-only rule
- D) A database design rule

**16. What is the diamond problem in OOP?**

- A) An ambiguity arising from multiple inheritance of the same base class
- B) A memory leak issue
- C) A sorting algorithm's worst case
- D) A type of exception

**17. What is the purpose of the 'super' keyword in Java?**

- A) To create a static method
- B) To refer to the parent class members
- C) To delete an object
- D) To define an interface

**18. Which OOP principle is primarily used to reduce code duplication by reusing existing classes?**

- A) Inheritance
- B) Polymorphism
- C) Abstraction
- D) Encapsulation

**19. Which of the following is true about a constructor?**

- A) It has a return type
- B) It is called automatically when an object is created
- C) It can be inherited like other methods
- D) It must be private

**20. What is object composition?**

- A) Building complex objects by combining simpler objects
- B) Inheriting from multiple classes
- C) Creating static objects
- D) Deleting objects

## DBMS & SQL

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**21. Which of the following best defines a stored procedure?**

- A) A precompiled collection of SQL statements stored in the database and executed as a unit
- B) A type of foreign key
- C) A column constraint
- D) A backup mechanism

**22. What is a view in SQL?**

- A) A physical copy of a table
- B) A virtual table based on the result of a SQL query
- C) A type of index
- D) A stored procedure

**23. What is the difference between WHERE and HAVING clauses?**

- A) WHERE filters rows before grouping, HAVING filters groups after aggregation
- B) They are identical in function
- C) HAVING is used only with SELECT \*
- D) WHERE cannot use comparison operators

**24. Which normal form eliminates transitive dependency?**

- A) 1NF
- B) 2NF
- C) 3NF
- D) 0NF

**25. Which SQL keyword is used to remove duplicate rows from a result set?**

- A) UNIQUE
- B) DISTINCT
- C) FILTER
- D) ONLY

**26. What is denormalization?**

- A) The process of adding redundancy to improve read performance
- B) Removing all keys from a table
- C) A type of join
- D) Encrypting a database

**27. Which type of JOIN returns all rows from the left table and matched rows from the right table?**

- A) INNER JOIN
- B) LEFT JOIN
- C) RIGHT JOIN
- D) CROSS JOIN

**28. What does the term 'referential integrity' ensure?**

- A) That foreign key values always match a primary key value in the referenced table (or are NULL)
- B) That all columns are indexed
- C) That all queries run faster
- D) That tables cannot be dropped

**29. Which SQL aggregate function returns the number of rows matching a condition?**

- A) SUM()
- B) COUNT()
- C) AVG()
- D) MAX()

**30. What is a self-join?**

- A) A join of a table with itself
- B) A join between two different databases
- C) A join without any condition
- D) An invalid SQL operation

**31. What is a primary key?**

- A) A column that can contain duplicate values
- B) A column (or set of columns) that uniquely identifies each row and cannot be NULL
- C) An index used only for sorting
- D) A key used only in foreign tables

**32. What is a composite key?**

- A) A key formed by combining two or more columns to uniquely identify a row
- B) A key that references another table
- C) An automatically generated key
- D) A key with only one column

**33. Which SQL command is used to change the structure of an existing table?**

- A) UPDATE
- B) ALTER
- C) MODIFY
- D) CHANGE

**34. What is a deadlock in the context of a database transaction?**

- A) Two or more transactions waiting indefinitely for resources locked by each other
- B) A syntax error in SQL
- C) A missing index
- D) A type of join

**35. Which of the following is true about a candidate key?**

- A) It can contain NULL values
- B) Every candidate key can be a primary key, but only one is chosen as such
- C) A table can have only one candidate key
- D) It must always be a foreign key

**36. Which SQL statement is used to combine the result-set of two or more SELECT statements with the same number of columns?**

- A) JOIN
- B) UNION
- C) MERGE
- D) GROUP BY

**37. What is the purpose of an index in a database?**

- A) To enforce referential integrity
- B) To speed up data retrieval operations
- C) To store backup data
- D) To encrypt table data

**38. What does ACID stand for in DBMS transactions?**

- A) Atomicity, Consistency, Isolation, Durability
- B) Accuracy, Consistency, Integration, Data
- C) Atomic, Complete, Independent, Durable
- D) Association, Consistency, Isolation, Data

**39. Which index type stores data rows in the same order as the index?**

- A) Non-clustered index
- B) Clustered index
- C) Bitmap index
- D) Hash index

**40. Which property of a transaction ensures that partial updates are never visible to other transactions?**

- A) Atomicity
- B) Consistency
- C) Isolation
- D) Durability

## Operating Systems

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**41. Which of these describes a 'zombie process'?**

- A) A process that has completed execution but still has an entry in the process table
- B) A process that never starts
- C) A process using 100% CPU
- D) A kernel-level thread

**42. Which of the following is an example of an inter-process communication (IPC) mechanism?**

- A) Pipes
- B) Compilers
- C) Loaders
- D) Linkers

**43. Which of these is a necessary condition to prevent deadlock by breaking 'circular wait'?**

- A) Impose a strict ordering of resource acquisition
- B) Allow unlimited resource requests
- C) Never release resources
- D) Increase the number of processes

**44. What is a monolithic kernel?**

- A) A kernel where all OS services run in kernel space as a single large process
- B) A kernel with no drivers
- C) A kernel used only in embedded systems
- D) A kernel that cannot be modified

**45. What is spooling?**

- A) Simultaneous Peripheral Operations On-Line - buffering data for devices like printers
- B) A type of deadlock
- C) A page replacement algorithm
- D) A CPU scheduling technique

**46. Which of the following is an example of external fragmentation?**

- A) Free memory scattered in small blocks between allocated partitions, none large enough for a new request
- B) Wasted space within an allocated fixed-size block
- C) A page fault
- D) A deadlock

**47. Which scheduling algorithm gives the shortest average waiting time (in theory, for a given batch)?**

- A) FCFS
- B) SJF (Shortest Job First)
- C) Round Robin with a large quantum
- D) Priority Scheduling

**48. What is the banker's algorithm used for?**

- A) Deadlock avoidance by checking safe states before allocating resources
- B) Memory compression
- C) File compression
- D) CPU scheduling

**49. What is a critical section in OS?**

- A) A section of code that accesses shared resources and must not be executed by more than one process/thread at a time
- B) A section reserved for the kernel only
- C) A type of page fault
- D) A memory segment for stack only

**50. What is the purpose of a context switch?**

- A) To save the state of a running process and load the state of another
- B) To terminate a process
- C) To allocate memory
- D) To format a disk

**51. What is the difference between paging and segmentation?**

- A) Paging divides memory into fixed-size blocks; segmentation divides memory into variable-sized logical units
- B) They are identical
- C) Segmentation is faster in all cases
- D) Paging is used only for disk storage

**52. What does the term 'fragmentation' refer to in memory management?**

- A) Wasted memory space due to allocation and deallocation patterns (internal or external)
- B) A type of process priority
- C) A CPU scheduling delay
- D) A file system backup issue

**53. Which of the following best describes multiprogramming?**

- A) Running multiple programs on multiple CPUs simultaneously only
- B) Keeping multiple programs in memory so the CPU always has one to execute, improving utilization
- C) Running only one program at a time
- D) A type of scheduling exclusive to real-time systems

**54. Which of the following is true about a microkernel architecture?**

- A) It moves most services to user space, keeping the kernel minimal
- B) It has no scheduler
- C) It cannot support IPC
- D) It is always slower with no exceptions

## Computer Networks

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**55. What is the function of a DNS server?**

- A) To translate human-readable domain names into IP addresses
- B) To assign private IP addresses within a LAN
- C) To encrypt HTTP traffic
- D) To manage email delivery

**56. What is the purpose of a subnet?**

- A) To divide a large network into smaller, more manageable segments
- B) To increase the number of MAC addresses
- C) To eliminate the need for routers
- D) To replace DNS

**57. Which of the following is true about IPv6 compared to IPv4?**

- A) IPv6 uses 32-bit addresses while IPv4 uses 128-bit
- B) IPv6 uses 128-bit addresses, providing a vastly larger address space than IPv4's 32-bit addresses
- C) IPv6 is identical to IPv4 in address length
- D) IPv4 has replaced IPv6 entirely

**58. Which of the following is a connectionless protocol?**

- A) TCP
- B) UDP
- C) FTP
- D) HTTP over TCP

**59. What is the primary purpose of a load balancer?**

- A) To distribute incoming network traffic across multiple servers to improve reliability and performance
- B) To encrypt data at rest
- C) To act as a firewall only
- D) To assign static IP addresses

**60. Which of the following best describes latency in networking?**

- A) The amount of data transferred per second
- B) The delay before data transfer begins following an instruction
- C) The number of devices on a network
- D) The physical distance between two devices

**61. What does a 'ping' command test?**

- A) Reachability and round-trip time to a host using ICMP echo requests
- B) DNS resolution only
- C) Bandwidth of a connection
- D) Server uptime history



**62. What is NAT (Network Address Translation) used for?**

- A) To translate private IP addresses to a public IP address for internet access
- B) To encrypt data
- C) To assign MAC addresses
- D) To resolve domain names

**63. Which protocol is used for secure remote login to a server?**

- A) Telnet
- B) SSH
- C) FTP
- D) SMTP

**64. Which of the following best describes a CDN (Content Delivery Network)?**

- A) A distributed network of servers that delivers content to users from a location geographically closer to them
- B) A type of firewall
- C) A DNS resolution protocol
- D) A database replication technique

**65. What does VPN stand for and what is its main purpose?**

- A) Virtual Private Network - creates a secure, encrypted connection over a public network
- B) Virtual Public Network - increases internet speed
- C) Verified Private Node - a type of router
- D) Virtual Protocol Number - an IP addressing scheme

**66. What is a MAC address?**

- A) A unique hardware identifier assigned to a network interface
- B) An IP address used only for routing
- C) A type of DNS record
- D) A session identifier

## Data Structures & Algorithms

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**67. What is the time complexity of inserting an element at the head of a singly linked list?**

- A)  $O(n)$
- B)  $O(1)$
- C)  $O(\log n)$
- D)  $O(n \log n)$

**68. What does BFS stand for in graph traversal?**

- A) Best First Search
- B) Breadth First Search
- C) Binary First Search
- D) Backtrack First Search

**69. What is a balanced binary search tree used to prevent?**

- A) Overflow errors
- B) Degeneration into a linked list (worst-case  $O(n)$  operations)
- C) Memory leaks
- D) Integer overflow

- 70. Which data structure is typically used to implement DFS (iteratively)?**
- A) Queue
  - B) Stack
  - C) Heap
  - D) Hash table
- 71. Which traversal of a Binary Search Tree gives elements in sorted order?**
- A) Pre-order
  - B) Post-order
  - C) In-order
  - D) Level-order
- 72. Which data structure would you use to check if a string of brackets is balanced?**
- A) Queue
  - B) Stack
  - C) Array
  - D) Linked List
- 73. Which data structure is best suited for implementing a priority queue efficiently?**
- A) Array
  - B) Linked List
  - C) Heap
  - D) Stack
- 74. What is the time complexity of Bubble Sort in the worst case?**
- A)  $O(n)$
  - B)  $O(n \log n)$
  - C)  $O(n^2)$
  - D)  $O(\log n)$
- 75. Which data structure is typically used to implement BFS?**
- A) Stack
  - B) Queue
  - C) Heap
  - D) Array only
- 76. What is a hash collision?**
- A) When two different keys map to the same hash bucket/index
  - B) When a hash table runs out of memory
  - C) When a key is deleted incorrectly
  - D) When two hash tables merge
- 77. Which of the following problems is a classic example solved using dynamic programming?**
- A) Binary search
  - B) 0/1 Knapsack problem
  - C) Bubble sort
  - D) Linear search

**78. Which sorting algorithm is considered stable and has  $O(n \log n)$  time complexity in all cases?**

- A) QuickSort
- B) Merge Sort
- C) Selection Sort
- D) Heap Sort

**79. What is the time complexity of Dijkstra's algorithm using a binary heap?**

- A)  $O(V^2)$
- B)  $O(E \log V)$
- C)  $O(V + E)$
- D)  $O(E^2)$

**80. What is dynamic programming primarily used for?**

- A) Solving problems by breaking them into overlapping subproblems and storing results to avoid recomputation
- B) Sorting arrays only
- C) Random number generation
- D) Compiling code faster

**81. Which of the following is a common technique to resolve hash collisions?**

- A) Chaining or open addressing
- B) Binary search
- C) Recursion
- D) Memoization

**82. What is the average time complexity of searching in a well-implemented hash table?**

- A)  $O(n)$
- B)  $O(\log n)$
- C)  $O(1)$
- D)  $O(n^2)$

**83. What is the space complexity of an iterative approach to computing Fibonacci numbers with memoization storing all  $n$  results?**

- A)  $O(1)$
- B)  $O(n)$
- C)  $O(n^2)$
- D)  $O(\log n)$

**84. Which algorithmic technique does Merge Sort primarily use?**

- A) Greedy
- B) Divide and Conquer
- C) Dynamic Programming
- D) Backtracking

## Programming Language Fundamentals

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**85. What does 'pass by reference' mean?**

- A) A copy of the variable's value is passed to the function
- B) The function receives a reference to the actual variable, so changes affect the original
- C) Variables cannot be passed to functions
- D) Only primitive types can be passed

**86. What is the difference between a compiler and an interpreter?**

- A) A compiler translates the whole program to machine code before execution, an interpreter executes code line-by-line
- B) They are identical
- C) Interpreters are always faster than compilers
- D) Compilers cannot detect syntax errors

**87. What is a lambda function/expression?**

- A) A named function stored in a file
- B) An anonymous, inline function typically used for short operations
- C) A type of class
- D) A compiler directive

**88. What is the purpose of a 'static' variable in a class (Java/C++)?**

- A) It belongs to the instance and is unique per object
- B) It belongs to the class rather than any instance and is shared across all instances
- C) It can only be used inside loops
- D) It is always private

**89. What is method chaining?**

- A) Calling multiple methods on the same object in a single statement by having each method return the object itself
- B) A way to declare multiple classes
- C) A type of exception handling
- D) A memory management technique

**90. Which keyword is used to handle exceptions in Python?**

- A) catch
- B) except
- C) rescue
- D) handle

**91. In Java, what is the difference between '==' and '.equals()' when comparing objects?**

- A) They always behave identically
- B) '==' compares references while '.equals()' compares content (if overridden)
- C) '.equals()' is only for primitives
- D) '==' is only used for Strings

**92. What is the purpose of the 'finally' block in exception handling (Java/Python)?**

- A) It executes only if an exception occurs
- B) It executes only if no exception occurs
- C) It always executes regardless of whether an exception occurred or not
- D) It is used to declare exceptions

**93. Which of the following best describes garbage collection?**

- A) A manual process where the developer frees memory explicitly
- B) An automatic process that reclaims memory occupied by objects no longer in use
- C) A compilation step
- D) A type of exception handling

**94. In C++, what does the 'const' keyword do when applied to a variable?**

- A) Makes the variable a pointer
- B) Prevents the variable's value from being modified after initialization
- C) Declares a static variable
- D) Declares a global variable

## Web & Cloud Fundamentals

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**95. What is the difference between IaaS, PaaS, and SaaS?**

- A) They are all the same cloud service model
- B) IaaS provides raw infrastructure, PaaS provides a platform to build apps, SaaS delivers ready-to-use software
- C) SaaS provides infrastructure while IaaS provides software
- D) PaaS is only for storage

**96. What is the purpose of a Content Delivery Network (CDN)?**

- A) To cache and serve content from servers geographically closer to users, reducing latency
- B) To encrypt all web traffic
- C) To replace DNS servers
- D) To manage database transactions

**97. What does 'statelessness' mean in the context of REST APIs?**

- A) The server stores client session state between requests
- B) Each request from a client contains all the information needed to process it, with no reliance on stored server-side session state
- C) The API has no state at all, including no data
- D) Stateless APIs cannot use authentication

**98. What does HTTP stand for?**

- A) HyperText Transfer Protocol
- B) HighText Transmission Protocol
- C) Hyperlink Transfer Text Protocol
- D) HyperText Trade Protocol

**99. What is the purpose of cookies in web applications?**

- A) To store small pieces of data on the client side to maintain state/session information
- B) To encrypt HTTPS traffic
- C) To speed up DNS resolution
- D) To compress images

**100. What is auto-scaling in cloud computing?**

- A) Automatically adjusting compute resources up or down based on demand
- B) Manually adding servers only
- C) A type of database indexing
- D) A billing model only

## Answer Key

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|      |      |      |      |      |      |      |      |      |       |
|------|------|------|------|------|------|------|------|------|-------|
| 1-A  | 2-A  | 3-B  | 4-A  | 5-B  | 6-C  | 7-A  | 8-A  | 9-B  | 10-A  |
| 11-A | 12-B | 13-B | 14-A | 15-A | 16-A | 17-B | 18-A | 19-B | 20-A  |
| 21-A | 22-B | 23-A | 24-C | 25-B | 26-A | 27-B | 28-A | 29-B | 30-A  |
| 31-B | 32-A | 33-B | 34-A | 35-B | 36-B | 37-B | 38-A | 39-B | 40-C  |
| 41-A | 42-A | 43-A | 44-A | 45-A | 46-A | 47-B | 48-A | 49-A | 50-A  |
| 51-A | 52-A | 53-B | 54-A | 55-A | 56-A | 57-B | 58-B | 59-A | 60-B  |
| 61-A | 62-A | 63-B | 64-A | 65-A | 66-A | 67-B | 68-B | 69-B | 70-B  |
| 71-C | 72-B | 73-C | 74-C | 75-B | 76-A | 77-B | 78-B | 79-B | 80-A  |
| 81-A | 82-C | 83-B | 84-B | 85-B | 86-A | 87-B | 88-B | 89-A | 90-B  |
| 91-B | 92-C | 93-B | 94-B | 95-B | 96-A | 97-B | 98-A | 99-A | 100-A |

Format: [Question No.]-[Correct Option Letter]. Content compiled from publicly available 2026 placement-prep resources and established CS-fundamentals question banks - verify current-year specifics before your interview.